LTE / LTE Advanced / LTE Adv. Pro - System Overview

Course objectives:

The LTE System Overview course offers a comprehensive – yet easily understandable – overview of the functionality of LTE or 4G. LTE, or strictly speaking the Evolved Packet System (EPS), consists of the evolved Radio Access Network (E-UTRAN) and the Evolved Packet Core (EPC) as defined in 3GPP technical specifications.

The course focuses on the network architecture, nodes and mode of operation of the entire LTE System. It accounts for interfaces, protocols and services provided by LTE, including Mobile Broadband, VoLTE and IoT. Message-level traffic cases are used throughout the course to demonstrate important functions and concepts.

Furthermore, the training introduces system enhancements added in 3GPP Releases 10 and 11, such as Carrier Aggregation, Enhanced MIMO, Relaying, Collaborative Multipoint and Heterogeneous Networks, forming LTE Advanced and LTE Advanced Pro.

Duration Instructional methods:

Option 1: On-Site Training: 4 days,
Classroom-style lecture using Power-point slides with interactive discussion,
Questions & Answers after each module.

Option 2: Virtual Classroom: 4 days total with 4 x 1.5 hours per day

Option 3: Virtual Classroom: 8 days total with 2 x 1.5 hours per day

Upon successful registration participants will receive dedicated LMS account credentials, allowing access to pre-training documents and eLearning material.
All participants will also receive comprehensive course materials.

Target Audience:

The course is addressed to technical and non-technical staff from the technology, marketing and management field, regulators and government officials, executives, requiring an overview about the LTE network concepts, architectures, standards, implementations and applications.

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Course Contents:

- Introduction to 5G
  - Motivations, goals and performance targets for LTE / LTE Adv / LTE Adv Pro
  - Standardization history from Release 8 to Release 11.

- EPS (Evolved Packet System) architecture, key components and their role
  - EPS architectural principles
  - Strata
  - EPS bearers and QoS
  - Integration with 2G (GSM) and 3G (UMTS)
  - Interfaces overview
  - Evolved Packet Core (EPC) functions
  - Long Term Evolution (LTE) functions

- The Evolved Packet Core (EPC)
  - EPC architecture
  - EPC nodes and functions
  - EPC interfaces

- The Evolved RAN (E-UTRAN)
  - RAN Functionality
  - OFDM / OFDMA / SC-OFDM Basics
  - Radio Interface Frequency Domain and Radio Interface Time Domain
  - RAN Frame Architecture
  - FDD vs TDD Modes
  - LTE Logical / Transport / Physical Channels and Signals
  - RAN Signaling Overview

- LTE MIMO Overview
  - LTE MIMO principles and schemes
  - MIMO evolution from Rel.8 to Rel.14

- Security in LTE Networks
  - EPS Security Architecture
  - User Equipment (UE) security aspects
  - Access Network security aspects
  - Core Network security aspects
  - Transport Network and IP security aspects

- EPS Signaling Protocols and selected End-to-End Interworking procedures
  - The signaling protocols stack
  - The network registration procedure
  - End-to-End connection establishment
  - Mobility procedures in LTE networks

- Voice Services with LTE - VoLTE
  - The role of IMS (IP Multimedia System)
  - SIP functionality overview
  - EPC and IMS architecture and node functions
  - VoLTE Service and Signaling examples

- Carrier Aggregation (CA)
  - Types of CA
  - Operating bands and bandwidth classes
  - Deployment scenarios
  - Serving cell types
  - Control channels in LTE Advanced
  - Cross carrier scheduling and mobility.

- Enhanced multiple antenna techniques
  - Overview of MIMO techniques
  - Transmission mode 9
  - Uplink MIMO
  - Reference signals in LTE Advanced
  - Enhanced uplink transmission scheme

- Relaying
  - Relay technology
  - Architecture
  - S1/X2 interfaces in relay operations
  - R-PDCCH – new control channel for relaying
  - Transmission timing for relay operations
  - Processes and procedures
  - Relay evolution in the future

- Collaborative Multipoint (CoMP)
  - CoMP technology basics
  - Joint Processing (JP)/Joint Transmission (JT)
  - Collaborative scheduling/beamforming
  - Dynamic Point Selection (DPS)
  - EPDCCH – new control channel
  - UE feedback and uplink CoMP transmissions.

- Heterogeneous Networks (HetNet)
  - Features of HetNet
  - Deployment challenges
  - Interference scenarios analysis
  - Interference suppression
  - Enhanced Inter Cell Interference Coordination (eICIC) technology
  - Backhaul role in HetNet

Pre-requisites:

Basic knowledge about mobile networks, IP networks and experience with IT infrastructure are beneficial.

Course Code: LTESysO

Price per student:

For Option 1, Classroom training : USD 2,400.00 per student

For Options 2, 3, Virtual Classroom training: USD 2,000.00 per student